Communications and Networks

Preliminaries

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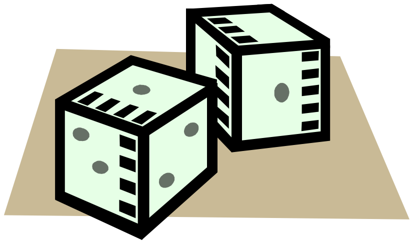
Outline

- Probability Theory
- Random (Stochastic) Process
- Fourier Transform

Discrete Random Variable

- The outcome of an experiment is from a finite or countable set S with *M* elements, use *X* to represent the outcome, then *X* is a discrete random variable



- The probability is defined:

$$P_i = \Pr(X = S_i), S_i \in \mathcal{S}$$

- Total probability:

$$\sum_{i=1}^{M} P_i = 1$$

Continuous Random Variable

- The outcome of an experiment is from a continuous set S, use X to represent the out come, then X is a continuous random variable
  - In general we define $\mathcal{S} = (-\infty, \infty)$
- First define the cumulative distribution function (CDF)

$$F(x) = \Pr(X \le x)$$

- Then define the probability density function (PDF)

$$p(x) = \frac{dF(x)}{dx} \quad \text{or} \quad F(x) = \int_{-\infty}^{x} p(u)du$$

Focus on continuous random variables in the following

Joint Distribution and Independence

- Two random variables *X* and *Y*, the joint distribution functions are defined as:

$$F(x,y) = \Pr(X \le x, Y \le y) = \int_{-\infty}^{x}\int_{-\infty}^{y} p(u,v)\,du\,dv$$

- They are independent if and only if

$$p(x,y) = p(x)p(y)$$

Conditional Distribution

- Define the conditional probability distribution function as

$$p(x|y) = \frac{p(x,y)}{p(y)}$$

- If X and Y are independent?

Statistical Averages of Random Variables

- The mean (expected value) of a R.V. *X*:

$$m_x = \mathbb{E}(X) = \int_{-\infty}^{\infty} x p(x) dx$$

- Variance of a R.V. *X* is:

$$\sigma_x^2 = \int_{-\infty}^{\infty} (x - m_x)^2 p(x) dx$$

- Equivalently:

$$\sigma_x^2 = \mathbb{E}(X^2) - [\mathbb{E}(X)^2] = \mathbb{E}(X^2) - m_x^2$$

Statistical Averages of Random Variables

- Consider two random variables *X* and *Y*,

$$\mathbb{E}(X+Y) =$$

$$\sigma^2_{X+Y} =$$

- If *X* and *Y* are independent?

Function of Random Variables

- Consider the functions of a group of R.V.s X_i

$$Y_i = g_i(X_1, X_2, ..., X_n)$$

- And the following inverse functions exist

$$X_i = g_i^{-1}(Y_1, Y_2, ..., Y_n)$$

- Then the joint PDF of Y_i is

$$\boxed{p_{\mathbf{Y}}(y_1,\cdots,y_n) = p_{\mathbf{X}}(x_1,\cdots,x_n)\,|J|}$$

$$\boxed{J = \begin{bmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_2}{\partial y_1} & \cdots & \frac{\partial x_n}{\partial y_1} \\ \frac{\partial x_1}{\partial y_2} & \frac{\partial x_2}{\partial y_2} & \cdots & \frac{\partial x_n}{\partial y_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_1}{\partial y_n} & \frac{\partial x_2}{\partial y_n} & \cdots & \frac{\partial x_n}{\partial y_n} \end{bmatrix}}$$

Function of Random Variables

- One-dimensional case:

$$Y = g(X)$$

- Then the joint PDF of *Y* is

$$p_Y(y) = p_X(x)\frac{dx}{dy} \qquad x = g^{-1}(y)$$

- The expectation of a function of R.V. *X*, i.e., *Y* is

$$m_y = \int_{-\infty}^{\infty} y p_Y(y) dy = \int_{-\infty}^{\infty} g(x) p_X(x) dx$$

Example: Gaussian Variable

- The PDF of a Gaussian R.V. is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_x} \exp\left[-\frac{(x-\mu_x)^2}{2\sigma_x^2}\right]$$

- The expectation of *X* is μ_x
- The variance of *X* is σ_x^2

- The sum of *n* independent Gaussian variables is still a Gaussian variable

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- Random (Stochastic) Process
- Fourier Transform

Random (Stochastic) Process

- A random variable indexed by the time parameter t: $X(t)$.
  - It is random at any time instance t

- We are interested in stationary processes
  - For any set of time instances, $t_1, t_2, \ldots, t_n$, and any time shift τ, the joint PDF satisfies

$$p(x(t_1), x(t_2), \ldots, x(t_n)) = p(x(t_1+\tau), x(t_2+\tau), \ldots, x(t_n+\tau))$$

Basically the process is homogeneous over time

Statistical Averages of a Random Process

- Mean of $X(t)$ is a function of time

$$m_{x(t)} = \mathbb{E}(X(t)) = \int_{-\infty}^{\infty} x(t)p(x(t))dx(t)$$

- But if $X(t)$ is stationary, the PDF is the same at any time t, we can drop the index t,

$$m_x = \mathbb{E}(X(t))$$

The mean is the same at any time, likewise for variance

- Define autocorrelation function (for stationary process), for any $t_1 \geq t_2$

$$\mathbb{E}(X(t_1)X(t_2)) = \phi(t_1, t_2) = \phi(t_1 - t_2) = \phi(\tau)$$

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Fourier Transform: Definition

- Fourier transform of a function *x*(*t*) (in general the function should have finite energy)

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$$

- Inverse Fourier transform

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi f t} df$$

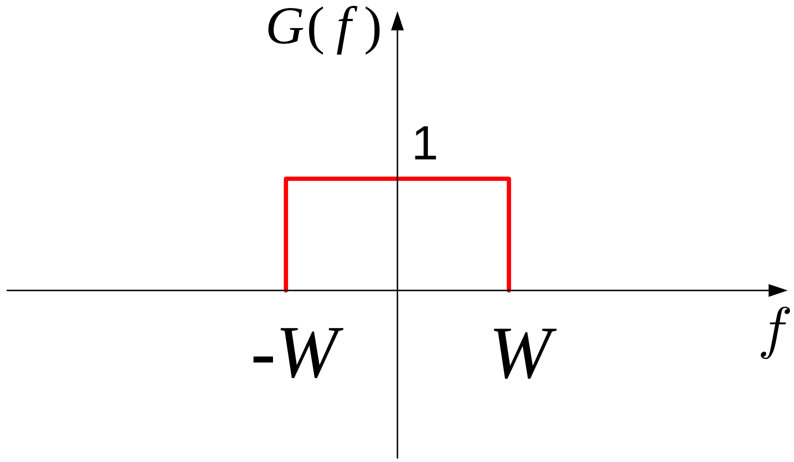
Properties of Fourier Transform

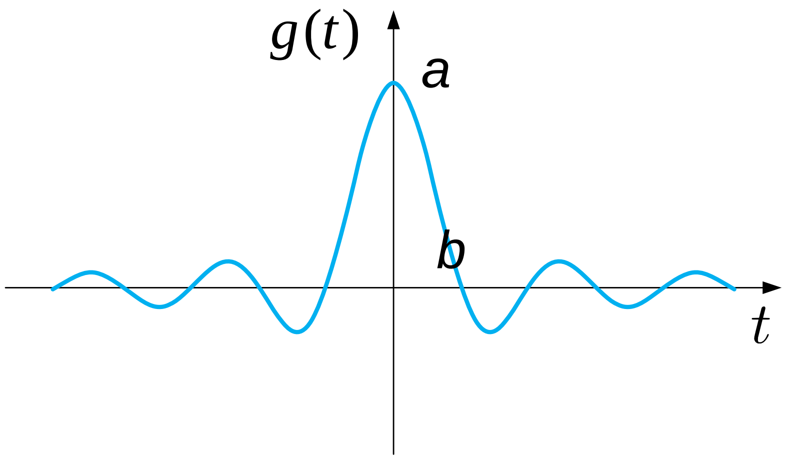
Time domain	Frequency domain	
$ax_1(t)+bx_2(t)$	$aX_1(f)+bX_2(f)$	**Linearity**
$x(t)$ is real	$X(-f)=X^*(f)$	**Conjugation**
$x(t-\tau)$	$X(f)e^{-j2\pi f\tau}$	**Time Shift**
$x(t)e^{j2\pi f_0 t}$	$X(f-f_0)$	**Frequency Shift**
$x(\frac{t}{T})$	$TX(Tf)$	**Scaling (T>0)**
$x_1(t)*x_2(t)$	$X_1(f)X_2(f)$	**Convolution**
$x_1(t)x_2(t)$	$X_1(f)*X_2(f)$	**Multiplication**
$\int_{-\infty}^{\infty}x(t)dt$	$G(0)$	

Useful Fourier Transform Pairs

Time domain $x(t)$	**Frequency domain** $X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$
$\delta(t)$	1
$\delta(t-\tau)$	$\exp(-j2\pi f\tau)$
$\delta_T(t) = \sum_{n=-\infty}^{+\infty} \delta(t-nT)$	$\frac{1}{T}\sum_{n=-\infty}^{+\infty} \delta(f-\frac{n}{T})$
$\cos(2\pi f_0 t)$	$\frac{1}{2}[\delta(f-f_0)+\delta(f+f_0)]$
$\mathrm{rect}(t) = \begin{cases} 1, \lvert t\rvert \le 0.5 \\ 0, \lvert t\rvert > 0.5 \end{cases}$	$\frac{\sin \pi f}{\pi f}$
$\mathrm{sinc}(t) = \frac{\sin \pi t}{\pi t}$	$\begin{cases} 1, \lvert f\rvert \le 0.5 \\ 0, \lvert f\rvert > 0.5 \end{cases}$

Example: Ideal Low-Pass Filter





$$G(f)=\begin{cases}1 & |f|\le W\\ 0 & |f|>W\end{cases}$$

$$g(t)=a\cdot\frac{\sin 2\pi Wt}{2\pi Wt}$$

$a = 2W$

$b = 1/2W$

Some Reading Recommendations

- Probability Theory (Haykin: Appendix 1)
- Random (Stochastic) Process (Haykin: 1.1-1.4)
- Fourier Transform (Haykin: Appendix 2.1 & 2.2)

Look forward to seeing you soon!